

Original Report

Metrics of success: Measuring impact of a departmental near-miss incident learning system



Matthew J. Nyflot PhD*, Jing Zeng MD, Aaron S. Kusano MD, SM, Avrey Novak BS, Thomas D. Mullen MD, PhD, Wendy Gao MD, Loucille Jordan RT(T), Patricia A. Sponseller MS, CMD, Joshua C. Carlson BS, Gabrielle Kane MB, EdD, FRCPC, Eric C. Ford PhD

Department of Radiation Oncology, University of Washington, Seattle, Washington

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Abstract

Purpose: There is a growing interest in the application of incident learning systems (ILS) to radiation oncology. The purpose of the present study is to define statistical metrics that may serve as benchmarks for successful operation of an incident learning system.

Methods and materials: A departmental safety and quality ILS was developed to monitor errors, near-miss events, and process improvement suggestions. Event reports were reviewed by a multiprofessional quality improvement committee. Events were scored by a near-miss risk index (NMRI) and categorized by event point of origination and discovery. Reporting trends were analyzed over a 2-year period, including total number and rates of events reported, users reporting, NMRI, and event origination and discovery.

Results: A total of 1897 reports were evaluated (1.0 reports/patient, 0.9 reports/unique treatment course). Participation in the ILS increased as demonstrated by total events (2.1 additional reports/month) and unique users (0.5 new users/month). Sixteen percent of reports had an NMRI of 0 (none), 42% had an NMRI of 1 (mild), 25% had an NMRI of 2 (moderate), 12% had an NMRI of 3 (severe), and 5% had an NMRI of 4 (critical). Event NMRI showed a significant decrease in the first 6 months (1.68–1.42, $P < .001$). Trends in origination and discovery of reports were broadly distributed between radiation therapy process steps and staff groups. The highest risk events originated in imaging for treatment planning (NMRI = 2.0 ± 1.1 ; $P < .0001$) and were detected in on-treatment quality management (NMRI = 1.7 ± 1.1 ; $P = .003$).

Conclusions: Over the initial 2-year period of ILS operation, rates of reporting increased, staff participation increased, and NMRI of reported events declined. These data mirror previously reported findings of improvement in safety culture endpoints. These metrics may be useful for other institutions seeking to create or evaluate their own ILS.

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* Corresponding author. University of Washington, 1959 NE Pacific St, Box 356043, Seattle, WA 98195–6043.

E-mail address: nyflot@uw.edu (M.J. Nyflot).

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Introduction

Incident learning describes the feedback loop in which near-miss events or errors are reported and then analyzed for root causes and contributing factors in order to inform interventions and process improvements. Incident learning systems (ILS) play an important role in safety for many industries¹⁻⁴ and interest in incident learning is growing in radiation oncology.⁵⁻¹¹ The importance of incident learning is further evidenced by the recently developed American Society for Radiation Oncology (ASTRO) practice accreditation standards that explicitly identify reporting and tracking of near-miss events as a part of a comprehensive culture of safety.¹² The ASTRO-American Association of Physicists in Medicine (AAPM) white paper *Safety is No Accident* states that “near-misses should be addressed with a similar vigor as that applied to errors.”¹³ A national ILS, the Radiation Oncology Incident Learning System (RO-ILS), has been developed by ASTRO and AAPM.

Despite this increased focus on incident learning, relatively little is known about the impact of ILS on improving patient safety or safety culture in radiation oncology. Numerous reports describing near-miss incident learning have appeared in the radiation oncology literature,⁵⁻¹¹ but it has been challenging to tie these to measureable impact. One recently reported endpoint, however, is safety culture with reported improvements in one institution following implementation of an ILS.¹⁴ Compared with baseline responses before use of incident learning, significant shifts in attitudes were observed, with individuals reporting increased engagement in safety improvement, increased awareness of safety issues, decreased concern of punitive actions, and increased confidence that their reports were being used to improve the system.

The purpose of the present study is to define statistical metrics that may serve as benchmarks for successful operation of an incident learning system. Established metrics that define “success” of incident learning systems in radiation oncology have yet to be defined, though evidence suggests that greater volume of event reporting reflects a better safety environment and thus fewer patient safety incidents.¹⁵ This theme is echoed in professional society recommendations that support high volumes of reporting as an indicator of positive safety culture^{16,17} and in institutional reports from radiation oncology departments that advocate for a high volume of reports.⁵ Similar metrics could be used to evaluate the success of incident learning systems for administrative, academic, or accreditation purposes.

In this report, we analyze data regarding trends in reporting over the initial 2-year period of operation of a departmental ILS. We present metrics demonstrating increased use of the system over time in terms of number of reports and number of participating users. Additionally, we present classification data describing broad usage over all steps of the radiation therapy workflow, including wide participation between staff groups. Finally, we define a

metric for near-miss event risk and demonstrate a reduction in average risk of reported events over time. These data may be useful for other institutions seeking to create or evaluate their own incident learning systems.

Methods and materials

A safety and quality incident learning system was developed to monitor departmental errors, near-miss events, and process improvement suggestions based on recommendations from the AAPM Working Group on the Prevention of Errors¹⁶⁻¹⁸ and with the approval of the University of Washington Institutional Review Board. Operational details of the system are published elsewhere¹⁴; briefly, staff are encouraged to report near-miss events or opportunities for quality improvement at any stage in the radiation therapy workflow. Reporting is accomplished via a software tool that is accessible on any departmental computer. Event reports are reviewed weekly by a multiprofessional quality improvement committee consisting of physicians, physicists, dosimetrists, therapists, residents, nurses, and other staff. Our institution has a hospital-wide reporting system for the reporting of events that reach the patients that is used in combination with the ILS.

Reports were categorized by the radiation therapy process step in which the event originated and was found. These process steps were derived from the AAPM consensus guidelines¹⁶ and consist of 8 areas of radiation therapy workflow separated into 103 individual process steps. Although it is recognized that incidents reaching the patient vary in severity, it must also be appreciated that near-miss events also vary in their potential risk for harm. We have therefore developed a near-miss risk index (NMRI) that scores near-miss events on a scale of 0 to 4 (none, mild, moderate, severe, critical, respectively).

The NMRI is an evaluation of the potential for an event to cause harm to patients. Events with fewer barriers for detection and greater clinical consequences are scored more highly. As such, the NMRI combines both severity and probability and can be best thought of as a risk score, a concept that has been similarly applied in areas such as failure mode and effects analysis. Note that the NMRI is not restricted to dosimetric consequences; events that do not directly involve irradiation are also evaluated. Examples include events that could result in pain, loss of consciousness, nonoptimal sedation or medication, or treatment delays. The NMRI is scored by consensus at the weekly multiprofessional review and is calculated for each process step in event point of origination and discovery. Table 1 shows the scoring schema and representative events.

Reporting trends were analyzed over time, including total number and rates of events reported, users reporting, NMRI, and the workflow point of incident origination and

Table 1 Definitions and representative examples for the near-miss risk index, which represents the potential risk of a near-miss event to harm a patient

Near-miss risk index	Criteria	Representative events
0	Event does not pose downstream risk in workflow.	Confusion during weekend treatment leading to a suggestion to add all phone numbers for weekend call team to department intranet. Phone numbers were scattered in multiple locations.
(No potential harm)	Event is not related to patient safety or quality of treatment.	Treatment plan documentation was incorrect for billing purposes. Dosimetrist was called to edit documentation.
1	Event may enhance the risk of other downstream errors.	Late planning volumes into dosimetry led to rushed planning and checks. Therapists could not fully complete film day checklist and physics check was performed off-hours.
(Mild potential harm)	Event may cause emotional distress or inconvenience to patient with no clinical impact.	Patient was not told to arrive early for intravenous contrast for simulation, therefore had to wait for the intravenous contrast. Extra time put simulation room behind schedule, delaying other patients and leading to rushed work by therapists.
2	Event enhances the risk of other critical downstream errors.	Wrong headrest used for one fraction of patient treatment, leading to suboptimal patient positioning.
(Moderate potential harm)	- Temporary pain or discomfort for patient. - Deviations from best practices, but with no obvious clinical impact.	Offsite physician scheduled for image guidance check. Patient was delayed on treatment table for 20 min in pain until another physician was found to approve cone beam computed tomography images.
3	Limited barriers to prevention of problem.	Linear accelerator backup jaws were not fully pushed behind the multileaf collimators in planning. Dose to surrounding tissue would have been increased by approximately 3%. Identified on pretreatment physics check.
(Severe potential harm)	Event with potential clinical impact that is noncritical.	Patient immobilization device deflated between treatments, requiring resimulation and replanning during treatment course. Patient missed a treatment day.
4	Extremely limited barriers to prevention of problem.	Patient receiving palliative radiation in one fraction was repositioned several times before treatment because of setup issues, and was almost treated at wrong source to surface distance.
(Critical potential harm)	Event with potentially critical clinical impact.	- Patient was set up and wrong leg was filmed for extremity treatment. Identified on port films before treatment delivery. - Patient receiving repeat radiation had planning volumes contoured on prior computed tomography data set, which were submitted to dosimetry for planning. Identified on pretreatment imaging.

discovery. Reports were excluded from analysis if they were blank, duplicated, retracted by the user, or erroneous. Trends were investigated per month and per 6-month interval. Total number of reports and unique users reporting versus months elapsed were fit with linear regression. Independent nonparametric Mann–Whitney tests were used to test changes in severity of reporting over 6-month intervals with a Bonferroni-corrected level of significance of $\alpha = 0.01$. Individual process steps were tested for significantly different risk indices versus all events with Mann–Whitney tests and Bonferroni-corrected $\alpha = 0.006$.

Results

Over the 2-year period between March 1, 2012, and February 28, 2014, 1995 event reports were accumulated. Ninety-eight reports met the exclusion criteria defined previously, resulting in a final dataset of 1897 reports. Averaged over the 2-year period, the number of reports per patient was 1.0 and the number of reports per unique treatment course was 0.9.

Participation in the ILS increased over time as demonstrated by total events reported per month (approximately 2.1 additional reports per month) and unique users reporting

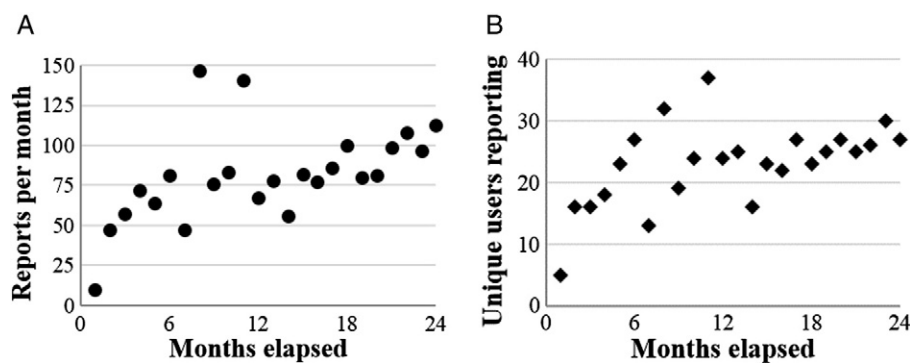


Figure 1 Increasing trends in monthly rate of reports (left) and unique users reporting (right) were observed over a 2-year period.

per month (0.5 additional users reporting per month) from a total of 87 clinical staff in the department (Fig 1). Although the average number of reports increased from approximately 50 to 100 reports per month, 2 months were outliers with nearly 150 events reported. Therapists, dosimetrists, and physicists reported the most events ($35\% \pm 13\%$, $29\% \pm 11\%$, and $24\% \pm 6\%$ per month, respectively; Fig 2). However, in any given month, the proportion of groups reporting was observed to vary widely. Fewer than 2% of events were reported anonymously.

Over all 1897 events, 16% of reports had an NMRI of 0 (none), 42% of reports had an NMRI of 1 (mild), 25% of reports had an NMRI of 2 (moderate), 12% of reports had an NMRI of 3 (severe), and 5% had an NMRI of 4 (critical). The NMRI showed a decreasing trend over the first 2 years (Fig 3). When investigating changes in the NMRI over 6-month intervals, a significant reduction was seen from months 1 to 6 to months 7 to 12 (average NMRI 1.68-1.42, $P < .001$ in unpaired t test), but no significant change was measured in the following 6-month intervals ($P = .74$, $P = .69$). Similar trends were observed when investigating only the highest risk events: 21% (64 of 303) of events had an NMRI of 3 to 4 in months 1 to 6,

whereas 15.9% (85 of 535), 15.2% (70 of 462), and 16.0% (93 of 583) had an NMRI of 3 to 4 in the following 6-month intervals.

Trends in origination and discovery of reports were broadly distributed between radiation therapy process steps, as depicted in Figure 4 and Table 2. The workflow step with the most reported events was treatment planning (33%), whereas the workflow step in which the most events were detected was treatment delivery (30%). The highest risk events originated in imaging for radiation therapy planning (NMRI = 2.0 ± 1.1 ; $P < .0001$). The highest risk events were detected in on-treatment quality management (NMRI = 1.7 ± 1.1 ; $P = .0007$), most commonly in weekly physics checks ($n = 78$), review of localization images, including cone beam computed tomography ($n = 34$), and review of portal images ($n = 22$). Events detected in treatment delivery also had a higher NMRI (1.6 ± 1.0 ; $P = .003$). The majority of reports that were not classifiable within these categories were designated as “other” (8.7% of reports for event origin, 2.7% of reports for event discovery; Table 2) and were for events that originated or were found outside of the radiation oncology department but reported by radiation oncology staff. (See Figure 4.)

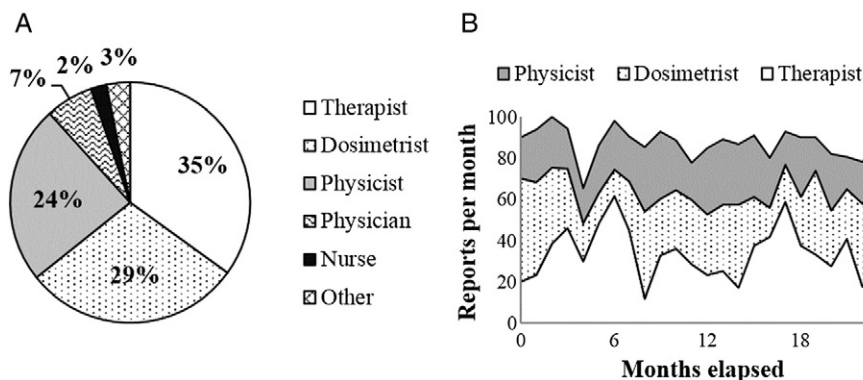


Figure 2 Rates of reporting by employment group showed a balanced distribution (left). When comparing rates of reporting of the 3 most prolific groups of reporters (right), large variability was observed month to month.

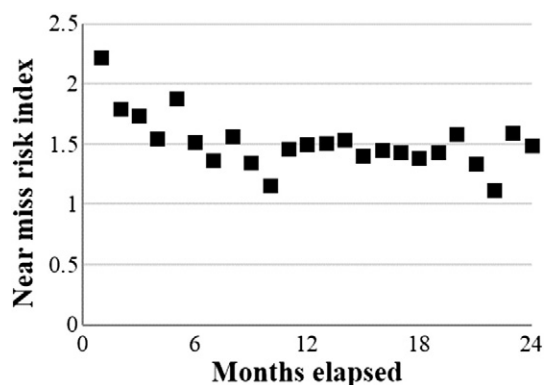


Figure 3 Mean near-miss risk index (NMRI) over time. When investigating changes in the NMRI over 6-month intervals, a significant reduction was seen from months 1-6 to months 7-12; however, no significant change was measured in the following 6-month intervals.

Discussion

Incident learning is a proven quality and safety improvement tool in high-risk, high-reliability industries such as air transportation and nuclear power and is gaining prominence in radiation oncology.¹⁻⁴ Support for the goal of high-volume reporting is provided by industrial theories such as Heinrich's Triangle and Reason's "Swiss cheese" model, which suggest that because near-misses and errors share similar root causes, reduction of near-misses will result in the reduction of errors.^{19,20} Recent support for ILS in radiation oncology includes endorsement from several professional societies, the recent launch of the RO-ILS in the United States, and inclusion into the recently developed ASTRO accreditation standards.

In this report, we analyzed data regarding participation and engagement regarding our departmental ILS. We observed a relatively high volume of near-miss events reported (1.0 per patient, approximately 50-100 reports per month). Additionally, increasing trends in reporting volume were observed following the introduction of the

ILS in terms of number of reports and number of unique users reporting. Furthermore, the presence of 2 outlier months in which nearly 150 events were reported appears to indicate that near-miss events are still somewhat underreported at our institution and continued increases in utilization of the ILS can be expected. These data differ from most previous reports on incident learning in the radiation oncology literature, which describe relatively low volumes of reporting (Table 3).

The result that rates of event reporting were seen to increase, not decline, following the introduction of the incident learning system is worthy of further reflection. Intuitively, one might expect that changes implemented in response to events would reduce the rate of reporting. We hypothesize that this effect was subsumed by the increase in number of users reporting and in their confidence that incident learning improves the safety and quality of care. These findings support the self-reported improvements in patient safety culture observed in our department following implementation of the incident learning system,¹⁴ such as reduction in perceived barriers to reporting and improved self-efficacy with more individuals valuing event reporting and having confidence that reports were contributing to safety improvement. Furthermore, we suggest that near-term increase in rates of reporting following implementation of the system should be used as a metric of success for ILS. This hypothesis is supported by data from the Agency for Healthcare Research and Quality (AHRQ) Hospital Survey on Patient Safety, in which higher rates of incident reporting were correlated with fewer safety incidents.¹⁵ We are further encouraged by our correlated result that the NMRI declined over time.

Similarly, wide participation in the ILS was observed, with 35% of reporting from treating or simulation therapists, 30% from dosimetrists, 24% from physicists, and additional participation from physicians, nurses, residents, and other staff. In contrast, previous reporting on utilization of ILS by staff role has been asymmetric; for example, the Radiation Oncology Safety Information System report⁹ indicates that 69% of reporting originated

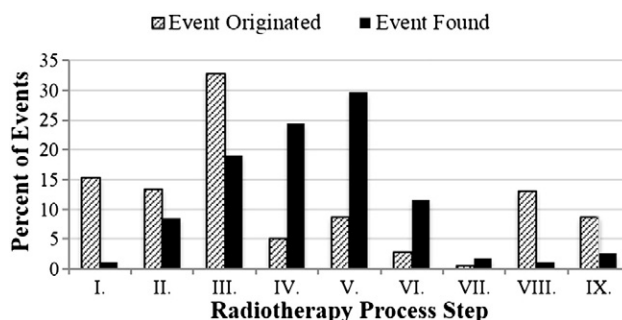


Figure 4 Graphical representation of trends in report origination and discovery as a function of radiation therapy process step. Reporting was broadly distributed with no more than one-third of events originating or discovered in a single step. Process step definitions are described in Table 2.

Table 2 Trends in report origination and discovery as grouped by the 8 radiation therapy process steps

Radiation therapy process step	Events originated	NMRI	Events found	NMRI
I: Patient assessment	292	1.5 ± 1.0	21	1.2 ± 0.9
II: Imaging for radiation therapy Planning	252	2.0 ± 1.1 ^a	161	1.4 ± 0.8
III: Treatment planning	621	1.5 ± 1.0	363	1.4 ± 1.0
IV: Pretreatment review and verification	96	1.4 ± 0.9	463	1.4 ± 1.0
V: Treatment delivery	163	1.5 ± 1.0	563	1.6 ± 1.0 ^a
VI: On-treatment quality management	54	1.4 ± 1.0	218	1.7 ± 1.1 ^a
VII: Posttreatment completion	8	0.5 ± 0.8	35	1.2 ± 1.2
VIII: Equipment and software quality management	246	1.2 ± 1.0 ^b	23	0.9 ± 1.0 ^b
IX: Other	165	1.0 ± 0.7 ^b	50	1.0 ± 1.1 ^b

NMRI, near-miss risk index.

^a Process steps that had significantly higher near-miss risk index versus all reports.

^b Steps with significantly lower NMRI versus all reports ($P < .006$).

from therapists at the treatment units, 11% from physicists, and 5% from dosimetrists. In addition, low rates of anonymous reporting were observed in the present study, and anonymous reporting is not encouraged in a just culture construct where obtaining more information about the incident is key.²¹ Broad participation between professional groups and low rates of anonymous are consistent with our prior analysis describing improvements in safety culture, such as decreased barriers to reporting and increased staff confidence in the value of incident learning.

Although it is recognized that incidents that reach the patient vary in severity, it must also be appreciated that near-miss events also vary in their potential risk for harm. For example, setting up and filming the wrong leg of a patient is more severe than an incorrectly labeled beam name. Both “errors” can contribute to actual incidents of harm but one is intuitively more severe than the other. The near-miss risk index was developed to prioritize analysis and process improvement resources, recognizing that traditional guidelines for error severity such as the

AHRQ Harm Scale are not generally useful for errors which do not reach the patient. Additionally, risk scoring is used in our clinic to prioritize which events are reported to the RO-ILS. Our investigation demonstrated a significant decrease in the average NMRI for reports over time. This decline may be attributed to process improvements as a result of reporting, increased comfort with reporting even minor events among department personnel, or other factors. A similar trend was shown in the study of incident reporting from Clark et al.,⁸ in which decrease in the number of actual high-severity incidents was described over a 4-year period, though the statistical significance of this trend was not reported and reporting rates of near-miss events also declined over this period.

Note that our near-miss risk index, from no potential harm (NMRI 0) to critical potential harm (NMRI 4), is similar to other near-miss severity scales,^{5,6,16} but differs in some respects. The suggested 0 to 10 point scales of the consensus recommendations proved challenging to use in practice, because much is unknown when evaluating the

Table 3 Incident learning systems from the literature that include near-miss events

Author, reference	Reports	Per year	Per patient	Per treatment course	Location	Dates gathered
Yeung (2005) ¹¹	624	60	0.05		Northeastern Ontario Regional Cancer Center, Canada	11/1992-12/2002
Arnold (2010) ⁷	688	230		0.18	Illawarra Cancer Centre, Australia	05/2004-05/2007
Bissonnette (2010) ¹⁰	1063	150		0.02	Princess Margaret Hospital, Canada	2001-2007
Cunningham 2010 ^d	1074	190	0.01		ROSIS (101 departments)	01/2003-08/2008
Mutic (2010) ⁹	2736	1370	0.60		Washington University, USA	08/2007-07/2009
Clark (2013) ⁵	2506	500			Ottawa Hospital, Canada	2007-2011
Terezakis (2013) ⁶	4407	1100	0.26		John Hopkins University and Washington University, USA	01/2007-12/2010
Present report	1897	950	1.0	0.86	University of Washington Medical Center, USA	03/2012-02/2014

Numbers in italics are not explicitly reported but estimated from available data in the report.

ROSIS, Radiation Oncology Safety Information System

potential severity of a near-miss event. Additionally, we found that framing potential severity only in terms of a single category (for example, dosimetric effects), was too limiting in a comprehensive safety and quality system. Furthermore, by moving to the concept of a risk score, our severity scoring system becomes more similar to other recommended safety and quality interventions such as failure mode and effects analysis. Our system is structurally most similar to that of Mutic et al., who score the severity of event from low (1) to critical (4). We also note that the full range of the NMRI scale is used in the near-miss data collected thus far because no NMRI category score has been assigned to fewer than 5% of events collected. We acknowledge that it is challenging to assess the potential severity of near-miss incidents, representing a potential source of bias. Determining the interobserver reliability of near-miss risk indices is a future direction of this work.

We found broad distribution of reporting in terms of event origination and discovery, with no more than one-third of reports coming from a single process step. In contrast, other investigators have reported less symmetric distribution of event reporting.^{8,9,11} This reinforces the data indicating wide participation between staff groups and correlating benefits to safety culture. Although the majority of events originated in patient assessment or treatment planning, the most severe events originated in imaging for radiation therapy planning, most commonly in the subcategories of documentation of patient positioning and setup, localization or marking the patient, and plan information transfers. Prominent discovery locations for severe events were at physics check, portal imaging, and cone beam computed tomography imaging, confirming the value of these processes that have been observed by others.^{22,23} More in-depth analysis of patterns of event origination and discovery is the subject of ongoing investigation. Treatment workflow categorization can be used to identify trends and prioritize improvement efforts. In contrast to the previously cited reports, event origination and discovery was classified based on the consensus recommendations for incident learning systems,¹⁶ which will aid in comparisons of multi-institutional data as well as analysis involving the national RO-ILS.

A limitation of the present work is that the results are not directly tied to patient outcomes. In practice, the link between safety and quality interventions and outcomes has proven elusive in radiation oncology outside of limited sets of data, such as the influence of radiation therapy plan quality on clinical trial outcomes.²⁴⁻²⁶ Outside of radiation oncology, however, data from the AHRQ Hospital Survey on Patient Safety has demonstrated that higher rates of event reporting lead to fewer safety incidents.¹⁵ An additional limitation is that incidence of actual radiation therapy errors is sufficiently low that tracking single-institution interventions is challenging, especially in light of the dynamic landscape of radiation therapy treatments and technologies. It is

incumbent on the radiation oncology community to design studies to track this data and demonstrate impact on patient outcome, a task that hopefully will be aided by the start of the national RO-ILS.

In summary, we find that over the initial 2-year period of operation of our departmental ILS, rates of reporting increased, staff participation in the system increased, and near-miss risk index of reported events declined. These data mirror previously reported findings of improved safety culture, reduced barriers to reporting, and increased staff confidence in the value of incident learning. Other metrics that may be considered indicative of a positive safety culture include a roughly equal distribution of event reporting between professional groups, equal distribution of event origination and discovery throughout the radiation therapy workflow, and a small proportion of reports logged anonymously. The metrics outlined here may be useful for other institutions seeking to create or evaluate their own incident learning systems or to develop tools that may be used to benchmark improvement in safety culture for administrative or accreditation purposes.

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